

News

Office Hours

Instructor - Mon 1-3

TA - Wed 12-2
Fri 1-3Animating in Matlab

- Review - Zero Length Spring
- Matlab Approach
- $\Rightarrow$ See Canvas

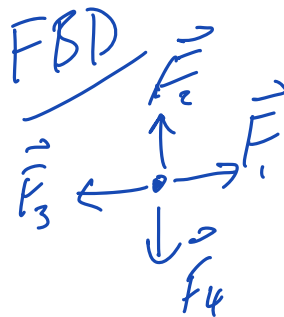
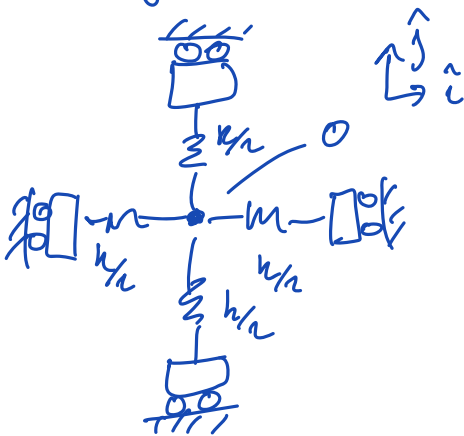
Assessment

Homework 1

Due Fri, 6pm.

Today Dr. van Paridon

will take it, Upson Conf. room, check Canvas

Analysis of Zero-Length Spring

$$\sum \vec{F}_i = \vec{F}_T$$

LMB $\vec{F}_T = m\vec{a}$

$$\hat{i} \Rightarrow m\ddot{x} = -kx$$

$$\hat{j} \Rightarrow m\ddot{y} = -ky$$

EOM (first order odes)

$$\dot{x} = v_x \quad \dot{v}_x = -\frac{k}{m}x$$

$$\dot{y} = v_y \quad \dot{v}_y = -\frac{k}{m}y$$

Define State Vector

$$X = \begin{bmatrix} x \\ y \\ v_x \\ v_y \end{bmatrix} = \begin{bmatrix} [r] \\ [v] \end{bmatrix}$$

$$\dot{X} = \begin{bmatrix} v_x \\ v_y \\ -k/m x \\ -k/m y \end{bmatrix} = \underbrace{f(X)}_{\text{myrhs}(t, X, \dots)}$$

Matlab Approach

